



Lagoon-Cell Prioritization in Closed-Loop Ash Disposal Systems Using an Integrated Fuzzy Analytic Hierarchy Process–Technique for Order Preference by Similarity to Ideal Solution Framework

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Abstract: Selecting suitable ash lagoon cells is a critical operational and environmental challenge in coal-fired power plant ash disposal systems. Conventional selection methods often emphasize disposal capacity while overlooking hydraulic connectivity, recirculation performance, operational flexibility, and infrastructure interdependencies. This study proposes an integrated Multi-Criteria Decision-Making (MCDM) framework combining the Fuzzy Analytic Hierarchy Process (FAHP) and Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) to prioritize feasible ash lagoon cells at the Poplar River Power Station (PRPS) under baseline and post-project conditions. Five criteria were evaluated: remaining capacity, hydraulic connectivity, discharge accessibility, recirculation impact, and operational flexibility. AHP was initially applied to assess judgment consistency, followed by FAHP to determine criteria weights under uncertainty. TOPSIS was then used to rank three lagoon-cell alternatives, and sensitivity analysis was conducted to assess robustness. Results identified remaining capacity as the most influential criterion. Cell 4E consistently ranked highest due to its balanced disposal capacity and operational performance, while the planned expansion of Cell 5 improved its post-project ranking. Sensitivity analysis confirmed ranking stability, demonstrating the robustness and practical applicability of the proposed FAHP–TOPSIS framework for infrastructure decision-making under uncertainty.

Keywords: Fuzzy Analytic Hierarchy Process; Technique for Order Preference by Similarity to Ideal Solution; Ash lagoon management; Multi-criteria decision making; Fly ash disposal

1 Introduction

Engineering decision-making problems frequently require the simultaneous consideration of multiple competing criteria, including technical feasibility, environmental impact, operational efficiency, and long-term sustainability. Single-factor approaches are generally inadequate for such problems, as they are unable to capture the multidimensional nature of real-world engineering systems. Multi-criteria decision-making (MCDM) methods have therefore emerged as a widely adopted analytical paradigm in engineering and environmental infrastructure research [1].

This multidimensional complexity is especially pronounced in waste management and environmental infrastructure planning, where decision problems rarely hinge on a single dominant criterion. Rather, multiple technical and operational considerations must be evaluated concurrently. Accordingly, MCDM techniques—including the Analytic Hierarchy Process (AHP), the Fuzzy Analytic Hierarchy Process (FAHP), and the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)—have been widely adopted for their ability to provide a structured and systematic basis for evaluating alternatives across several criteria simultaneously [2, 3].

Recent advances in decision science have further demonstrated that many engineering decisions are not only multi-criteria in nature but also inherently uncertain and dynamic. In practice, certain criteria resist precise quantification and must instead be assessed through engineering judgment or linguistic preference scales. This

reality has driven growing interest in hybrid decision-making frameworks that combine robust criteria-weighting methods with systematic ranking techniques. FAHP has been widely employed to capture and propagate uncertainty in pairwise comparisons, while TOPSIS is commonly applied to rank alternatives based on their relative proximity to ideal and anti-ideal solutions [4–6].

Despite the broad application of MCDM methods in environmental engineering, the preponderance of existing studies addresses greenfield site-selection problems, including landfill siting, wastewater treatment facility location, and waste-to-energy plant planning. While such studies offer valuable methodological contributions, they generally treat alternatives as spatially and operationally independent entities. Comparatively limited attention has been directed toward operational decision-making within existing interconnected infrastructure systems, where alternatives are subject to shared hydraulic behaviors, operational constraints, and system-level interdependencies [7, 8].

This gap is particularly consequential in the context of coal-fired power plant ash disposal systems. In wet ash management operations, ash slurry is conveyed into lagoon cells where solids settle and water is retained or recirculated for reuse within the facility. Ash lagoons thus serve not merely as passive storage structures but as functionally integrated components of a broader ash disposal and water recirculation system. Accordingly, lagoon-cell selection decisions must account for both individual storage capacity and system-level operational performance characteristics [9–11].

This operational complexity is directly applicable to the ash lagoon system at the Poplar River Power Station (PRPS), which operates as a closed-loop network of interconnected lagoon cells. Individual cells vary with respect to hydraulic connectivity, discharge accessibility, recirculation influence, and operational flexibility. A cell with substantial remaining capacity may nonetheless prove operationally less suitable due to poor hydraulic integration with the broader system, while a hydraulically favorable cell may lack adequate disposal capacity to meet operational demands. Lagoon-cell selection at PRPS therefore cannot be reduced to a capacity-based criterion alone.

To address this challenge, the present study develops and applies a structured MCDM framework to evaluate and prioritize three feasible lagoon-cell alternatives at PRPS. Five criteria are considered: remaining capacity, hydraulic connectivity, discharge accessibility, recirculation impact, and operational flexibility. Both baseline and post-project operational scenarios are examined to assess the influence of planned infrastructure modifications on lagoon-cell suitability rankings.

The main objectives of this study are as follows:

1. To evaluate and prioritize feasible ash lagoon cell alternatives at PRPS using an integrated FAHP–TOPSIS MCDM framework.
2. To determine the relative importance of key operational and engineering criteria while validating the consistency of expert judgments.
3. To compare the suitability of lagoon-cell alternatives under baseline and post-project capacity scenarios.
4. To develop a robust decision-support framework for ash lagoon management in interconnected ash disposal systems under uncertainty.

The primary research gap motivating this study is the limited application of MCDM approaches to operational lagoon-cell selection within closed-loop ash disposal systems. Existing studies have focused predominantly on greenfield facility siting, with comparatively little attention directed toward the prioritization of alternatives within existing interconnected lagoon infrastructure. Furthermore, many published MCDM frameworks operate under static decision environments and do not account for anticipated infrastructure modifications or evolving capacity conditions—factors that are integral to the long-term management of operational ash disposal systems. The present study addresses these limitations by applying an integrated FAHP–TOPSIS framework to the PRPS lagoon system under both baseline and post-project operational scenarios.

2 Literature Review

2.1 Ash Lagoon Management and Operational Decision Challenges

Ash lagoons play a critical role in the disposal of coal combustion by-products and require effective management to minimize environmental and operational risks [12]. Previous studies have primarily focused on environmental impacts, restoration strategies, and rehabilitation of fly ash disposal sites (Table 1). Research has examined vegetation-based stabilization, phytoremediation, ecological restoration, and biodiversity conservation within fly ash lagoons [11, 13, 14]. These studies highlight challenges associated with erosion, leachate generation, poor substrate quality, and long-term environmental sustainability [9].

While these investigations provide valuable insights into the environmental management of ash disposal facilities, they largely focus on post-disposal rehabilitation rather than operational decision-making. In practice, power plants often operate interconnected lagoon systems where the selection of disposal cells affects hydraulic performance, recirculation efficiency, infrastructure utilization, and future expansion planning. Consequently, operational lagoon-cell prioritization requires a broader evaluation framework that considers both engineering and environmental factors.

Table 1. Summary of relevant literature

Reference	Objective	Country	Focus Area	Method/Tool
Mvembe et al. [15]	Applied an integrated multi method decision framework for waste management evaluation	Africa	Waste management decision making	AHP + TOPSIS + SWARA
Abdelmagid et al. [16]	Identified suitable locations for a wastewater treatment plant using multiple spatial and environmental criteria	Iraq	Wastewater treatment plant site selection	FAHP+GIS
Ahmed et al. [4]	Applied a hybrid FAHP-TOPSIS approach to landfill site selection	Canada	Landfill site selection	FAHP + TOPSIS
Kolar et al. [14]	Investigated biodiversity and conservation potential of freshwater habitats in fly ash sedimentation lagoons	Czech Republic	Fly ash sedimentation lagoons	Field ecological assessment
Sambiani et al. [8]	Identified suitable locations for waste-to-energy facilities and optimized site allocation using GIS-based FAHP and location allocation analysis	Togo	Waste-to-energy facility siting	GIS-based FAHP + Location-allocation
Maiti and Jaiswal [11]	Reviewed restoration constraints and integrated restoration approaches for fly ash disposal area	India	Ecological restoration of fly ash disposal area	Field sampling and bioaccumulation assessment
Pasalari et al. [7]	Selected landfill sites using environment, technical and social criteria under uncertainty	Iran	Landfill site selection	AHP-Fuzzy GIS
Sun et al. [17]	Reviewed the management system, disposal methods and utilization of municipal solid waste incineration fly ash in the United States	United States	MSW incineration fly ash management	Review
Beskese et al. [18]	Evaluated landfill alternatives using fuzzy weighting and ranking techniques	Turkey	Landfill site selection	Fuzzy AHP, Fuzzy TOPSIS
Mardani et al. [1]	Reviewed the development and application of major MCDM methods across engineering, management and environmental decision making	General	MCDM methods and applications	Review
Pandey [13]	Examined the phytostabilization potential of <i>Ricinus communis</i> on fly ash disposal sites	India	Fly ash disposal site phytoremediation	Field study
Haynes [9]	Reviewed long term management needs for fly ash disposal and the environmental challenges	International	Fly ash disposal site management	Review
Maiti et al. [10]	Assessed metal bioaccumulation and translocation in natural vegetation growing on fly ash lagoons	India	Fly ash lagoon vegetation and bioremediation	Field Study

Note: AHP = Analytic Hierarchy Process; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; SWARA = Step-wise Weight Assessment Ratio Analysis; FAHP = Fuzzy Analytic Hierarchy Process; GIS = Geographic Information System.

2.2 Multi-Criteria Decision-Making Applications in Environmental Infrastructure

The MCDM methods are widely used to address complex engineering problems involving multiple and often conflicting criteria [1]. In environmental infrastructure planning, MCDM approaches have been extensively applied to landfill siting, wastewater treatment plant selection, waste-to-energy facility planning, and other waste-management decisions [4, 7, 8, 16, 18].

Among available techniques, hybrid approaches combining criteria-weighting and alternative-ranking methods

have gained significant attention. These studies demonstrate that MCDM frameworks effectively integrate technical, environmental, economic, and social considerations under uncertainty. However, most applications focus on greenfield site-selection problems where alternatives are spatially independent and evaluated primarily through location-based criteria. Comparatively fewer studies address operational decision-making within existing infrastructure systems characterized by physical and hydraulic interdependencies among alternatives.

2.3 Fuzzy Analytic Hierarchy Process–Technique for Order Preference by Similarity to Ideal Solution for Decision-Making under Uncertainty

AHP remains one of the most widely adopted methods for structuring decision problems and deriving criteria weights through pairwise comparisons [2]. A major advantage of AHP is its ability to assess judgment consistency using the Consistency Ratio. However, conventional AHP assumes precise judgments and may not adequately capture uncertainty inherent in engineering decision-making.

To overcome this limitation, the FAHP incorporates fuzzy set theory to represent linguistic assessments and uncertain expert preferences. Once criteria weights are established, the TOPSIS provides a transparent and computationally efficient method for ranking alternatives based on their relative distances from ideal and non-ideal solutions [6, 19].

The FAHP–TOPSIS combination has been successfully applied in environmental and infrastructure planning because it simultaneously addresses uncertainty in criteria weighting and provides a systematic ranking of alternatives. Its ability to integrate expert judgment with quantitative evaluation makes it particularly suitable for complex operational decisions involving multiple engineering criteria.

2.4 Research Gap and Contribution

Despite extensive applications of MCDM methods in environmental infrastructure planning, several important gaps remain. First, existing studies predominantly focus on greenfield site-selection problems in which alternatives are spatially independent. Limited attention has been given to operational decision-making within existing interconnected infrastructure systems. Second, few studies explicitly consider issues arising from hydraulic connectivity, recirculation performance, and infrastructure constraints. Third, scenario-based assessments that account for future capacity expansions and changing operational conditions remain limited. Finally, ash lagoon systems are largely absent from the MCDM literature, with prior research focusing primarily on environmental restoration and rehabilitation rather than operational prioritization.

This study addresses these gaps by developing an integrated FAHP–TOPSIS framework for evaluating ash lagoon cells within the interconnected ash disposal network of the PRPS. The framework incorporates operational, hydraulic, and capacity-related criteria while evaluating alternatives under both baseline and post-project scenarios. By extending MCDM applications beyond conventional site-selection problems, the study provides a practical decision-support approach for managing complex infrastructure systems under uncertainty.

3 Methodology

3.1 Research Framework

This study employed an integrated MCDM framework combining the FAHP and the TOPSIS to identify the most suitable ash lagoon cell for future ash slurry disposal at PRPS under both baseline and post-project conditions. The methodology consisted of four stages. First, the decision problem was formulated by identifying feasible lagoon-cell alternatives and evaluation criteria. Second, FAHP was applied to derive criteria weights from fuzzy pairwise comparison matrices. Third, the obtained weights were incorporated into the TOPSIS model to rank the alternatives according to their proximity to the ideal solution. Finally, the rankings were compared under baseline and post-project capacity scenarios to assess the impact of capacity expansion on lagoon-cell suitability.

3.2 Fuzzy Analytic Hierarchy Process for Criteria Weighting

Fuzzy AHP is an effective method for determining the relative importance of evaluation criteria while incorporating the uncertainty inherent in subjective human judgments. To capture this ambiguity and imprecision, fuzzy numbers are employed to convert linguistic assessments into quantitative measures, enabling a more realistic and reliable representation of expert judgments within the decision-making process.

Let the fuzzy comparison matrix is defined as:

$$\tilde{A} = [\tilde{a}_{ij}] \quad (1)$$

where, elements are triangular fuzzy numbers represented as follows:

$$\tilde{a}_{ij} = (l_{ij}, m_{ij}, u_{ij}) \quad (2)$$

where, l , m and u are the lower, middle and upper values of the fuzzy judgment, respectively. According to the fuzzy logic, reciprocal comparison is described as follows:

$$\tilde{a}_{ji} = \left(\frac{1}{u_{ij}}, \frac{1}{m_{ij}}, \frac{1}{l_{ij}} \right) \quad (3)$$

Then, the fuzzy AHP model requires constructing the fuzzy pairwise comparison matrix using the triangular fuzzy scale presented in Table 2. Fuzzy weighting calculations were carried out according to Buckley [3] and FAHP technique proposed by Abdelmagid et al. [16].

Table 2. Triangular fuzzy numbers scale used in this study, adapted from [16]

Crisp Value	Linguistic Meaning	Triangular Fuzzy Number
1	Equal importance	(1, 1, 1)
2	Slight to moderate importance	(1, 2, 3)
3	Moderate importance	(2, 3, 4)
4	Moderate to strong importance	(3, 4, 5)
5	Strong importance	(4, 5, 6)

The calculation of the fuzzy geometric mean for criterion $\tilde{r}_i$ proceeds as follows:

$$\tilde{r}_i = \left(\prod_{j=1}^n \tilde{a}_{ij} \right)^{1/n} \quad (4)$$

where, n is the number of criteria. Next, the fuzzy weight of criterion $\tilde{w}_i$ can be computed as:

$$\tilde{w}_i = \tilde{r}_i \otimes \left(\sum_{i=1}^n \tilde{r}_i \right)^{-1} \quad (5)$$

To make criteria comparable, defuzzification should be performed using the centroid method as follows:

$$M_i = \frac{l_i + m_i + u_i}{3} \quad (6)$$

After that, the normalized non-fuzzy weight can be computed as:

$$N_i = \frac{M_i}{\sum_{i=1}^n M_i} \quad (7)$$

These weights were later used for ranking alternatives via the TOPSIS technique.

3.3 Crisp Analytic Hierarchy Process Consistency Check

Although fuzzy pairwise comparisons were employed in the analysis, it is essential to verify the consistency of expert judgments to ensure that the decision-maker's evaluations are logically coherent and reliable. Therefore, a conventional (crisp) AHP consistency test was conducted before applying the Fuzzy AHP procedure to assess the consistency of the pairwise comparison matrix. This step is necessary because consistency in the original crisp judgments does not automatically guarantee consistency in the corresponding fuzzy comparison matrix, even though the latter is derived from the same preference structure.

The consistency index CI was calculated as:

$$CI = \frac{\lambda_{\max} - n}{n - 1} \quad (8)$$

where, $\lambda_{\max}$ is the largest eigenvalue of the pairwise comparison matrix and n is the number of criteria. Further, the consistency ratio CR is calculated as:

$$CR = \frac{CI}{RI} \quad (9)$$

where, RI is the random index corresponding to the matrix size. In accordance with AHP theory, consistency was deemed acceptable if $CR < 0.10$ [2].

3.4 Technique for Order Preference by Similarity to Ideal Solution for Ranking Lagoon Alternatives

Once the weights were obtained, the next step was to apply the TOPSIS method for ranking alternatives. TOPSIS seeks to choose the optimal alternative by maximizing its distance to the negative ideal solution and minimizing distance to the positive ideal solution [6, 19].

Let x_{ij} be the performance of alternative i relative to criterion j . Then the normalized decision matrix is defined as follows:

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}} \quad (10)$$

where, m is the number of alternatives.

Further, the weights of criteria have to be multiplied by their normalized values:

$$v_{ij} = w_j r_{ij} \quad (11)$$

where, w_j is the normalized weight of criterion j .

The positive ideal solution (A^+) is:

$$A^+ = \{v_1^+, v_2^+, \dots, v_n^+\} \quad (12)$$

The negative ideal solution (A^-) is:

$$A^- = \{v_1^-, v_2^-, \dots, v_n^-\} \quad (13)$$

The separation distance from the positive ideal solution is:

$$S_i^+ = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^+)^2} \quad (14)$$

The separation distance from the negative ideal solution is:

$$S_i^- = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^-)^2} \quad (15)$$

The closeness coefficient is:

$$CC_i = \frac{S_i^-}{S_i^+ + S_i^-} \quad (16)$$

The best alternative is identified as having the highest closeness coefficient CC_i .

4 Application of Framework

This section describes the step-by-step implementation of the proposed framework.

4.1 Study Area and Decision Context

The case study examines the ash lagoon system at PRPS, Saskatchewan. The lagoon operates as an interconnected multi-cell ash disposal system in which ash slurry is discharged into a designated disposal cell. Following sedimentation of ash particles within the disposal area, clarified water is conveyed through a series of downstream cells before being recirculated back to the power station. Consequently, the lagoon system performs not only a storage function but also plays a critical role in the overall ash disposal, water treatment, and recirculation process. Therefore, the selection of an appropriate disposal cell must be evaluated from both storage capacity and system functionality perspectives.

This study addresses the decision problem of identifying the most suitable lagoon cell for future ash slurry disposal under both baseline and post-project operating conditions. The problem extends beyond a simple capacity allocation exercise, as the suitability of a disposal cell is influenced by multiple engineering and operational considerations. A cell with substantial remaining capacity may not necessarily represent the optimal alternative if it adversely affects operational efficiency, hydraulic connectivity, or recirculation performance within the lagoon system. Accordingly, the evaluation of candidate cells is formulated as a multi-criteria engineering decision-making problem that incorporates both capacity-related and system-level performance factors.

4.2 Selection of Alternatives

Three lagoon cells-Cell 4E, Cell 4W, and Cell 5-were identified as feasible disposal alternatives because they represent the realistic operational options available within the PRPS lagoon system. The objective of this study is not to assess all lagoon areas but rather to evaluate those cells that can be practically utilized for future ash slurry disposal operations.

Cell 4E is characterized by the largest available disposal capacity under baseline conditions and favorable discharge accessibility, making it a strong candidate for disposal activities. Although Cell 4W has a comparatively lower remaining capacity, it remains a viable alternative due to its strong hydraulic connectivity within the lagoon network, which supports efficient system operation. In contrast, Cell 5 exhibits lower suitability under baseline conditions because of its limited disposal capacity. However, following the planned capacity enhancement project, the available storage volume of Cell 5 increases substantially, improving its overall suitability and positioning it as a competitive disposal alternative under future operating conditions.

4.3 Selection and Justification of Criteria

Five evaluation criteria were identified to assess the suitability of the selected lagoon-cell alternatives, as they collectively capture the key engineering and operational factors influencing future ash slurry disposal at PRPS (Table 3).

Table 3. Criteria definition and engineering meaning

Criterion	Description	Engineering Meaning
Remaining Capacity (C1)	Normalized volume available for disposal in the selected case study	Indicates the amount of time that the cell could continue receiving ash without becoming operationally restricted
Hydraulic Connectivity (C2)	How suitable the cell is, from a hydraulic standpoint	Indicates how easy it would be for the cell to become incorporated in the current hydraulic pathway
Discharge Accessibility (C3)	Relative ease of discharge to the cell	Indicates the ease of directing ash slurry to the selected cell
Recirculation Impact (C4)	Relative effect of the selected option on water recirculation	Indicates how efficiently selected cell affects recirculation
Operational Flexibility (C5)	Operational flexibility associated with using the cell	Indicates the degree of operational flexibility associated with the cell

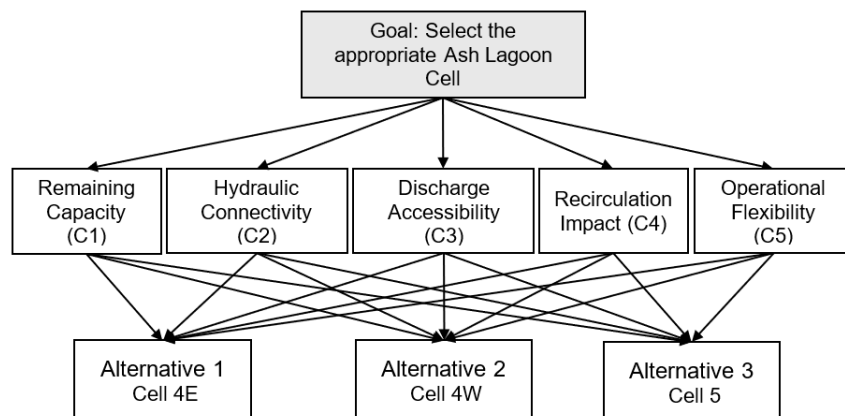


Figure 1. Analytic Hierarchy Process (AHP) structure

Remaining capacity (C1) was selected because it directly determines the operational lifespan of a disposal cell before reaching its storage limit. Given that the primary function of the lagoon system is ash slurry containment and disposal, this criterion is of fundamental importance.

Hydraulic connectivity (C2) was included because the lagoon operates as an integrated hydraulic network. Effective connectivity between cells is essential to maintain efficient slurry transport, sedimentation processes, and overall system performance.

Discharge accessibility (C3) reflects the practicality of delivering ash slurry to a disposal cell. A cell may exhibit favorable hydraulic characteristics yet present operational challenges for slurry discharge. Therefore, accessibility must be considered to ensure feasible and efficient disposal operations.

Recirculation impact (C4) evaluates the influence of a disposal cell on water-flow behavior within the recirculation system. As the PRPS lagoon operates in a closed-loop configuration, changes in recirculation performance can directly affect plant operations. Consequently, this criterion represents a critical system-level consideration.

Operational flexibility (C5) measures the extent to which a disposal cell supports adaptable and efficient operation under varying operational conditions. Although it may be less influential than storage capacity, operational flexibility remains an important factor for long-term planning and system management.

Together, these five criteria provide a comprehensive framework for evaluating lagoon-cell alternatives by considering both disposal capacity and the broader operational performance of the interconnected lagoon system. The decision hierarchy structure of this study is presented in Figure 1.

4.4 Expert Input for Qualitative Assessment

Several evaluation criteria considered in this study, particularly hydraulic connectivity, discharge accessibility, recirculation impact, and operational flexibility, are qualitative in nature and cannot be adequately characterized using numerical data alone. Their assessment therefore relied on expert engineering judgment and practical knowledge of the existing PRPS ash lagoon system.

To enhance the reliability and credibility of the qualitative evaluations, the assessment process was informed through consultations with experienced plant personnel, including plant engineers, operations specialists, ash-handling operators, and technologists who possess extensive knowledge of lagoon planning, operation, and maintenance. Their expertise contributed to the development of the pairwise comparison judgments used in the weighting process and supported the evaluation of alternatives for criteria where operational insight was critical. By incorporating practitioner knowledge into the decision-making framework, the analysis better reflects the practical realities and operational constraints associated with lagoon-cell selection.

4.5 Data Inputs and Decision Matrix Development

The input data used for analysis include quantitative information (the remaining capacity of lagoon cells) and qualitative information (hydraulic connectivity, discharge access, impact of recirculation and operational flexibility). The quantitative component was represented as remaining capacities of lagoon cells measured under two scenarios: (i) baseline, and (ii) post-project condition where Cell 5 has been enlarged. This variable stands for capacity.

On the contrary, qualitative attributes are represented as normalized numerical values between 0 and 1 and reflect engineering judgment, plant layout and operation of the lagoon system. Altogether, two decision matrices were generated for further application of TOPSIS.

4.6 Determination of Criteria Weights

4.6.1 Fuzzy pairwise comparison matrix and criteria weights

Based on expert feedback, the relative importance of the criteria was expressed using triangular fuzzy numbers to account for uncertainty in the decision-making process. The fuzzy pairwise comparison matrix and the corresponding criteria weights obtained using Buckley's Fuzzy AHP method are presented in Table 4. The results indicate that remaining capacity is the most influential criterion, followed by hydraulic connectivity, recirculation impact, discharge accessibility, and operational flexibility. These results highlight the dominant role of capacity-related considerations in ash lagoon selection while confirming the importance of hydraulic and operational factors.

Table 4. Fuzzy pairwise comparison matrix

Criterion	C1	C2	C3	C4	C5	Weight
C1	(1,1,1)	(2,3,4)	(4,5,6)	(3,4,5)	(4,5,6)	0.471
C2	(0.25,0.33,0.5)	(1,1,1)	(1,2,3)	(1,2,3)	(2,3,4)	0.205
C3	(0.17,0.2,0.25)	(0.33,0.5,1)	(1,1,1)	(0.33,0.5,1)	(1,2,3)	0.105
C4	(0.2,0.25,0.33)	(0.33,0.5,1)	(1,2,3)	(1,1,1)	(2,3,4)	0.151
C5	(0.17,0.2,0.25)	(0.25,0.33,0.5)	(0.33,0.5,1)	(0.25,0.33,0.5)	(1,1,1)	0.067

4.6.2 Defuzzification and consistency check

To verify the reliability of the expert judgments, the fuzzy comparison matrix was defuzzified to generate the corresponding crisp AHP pairwise comparison matrix shown in Table 5. The consistency of the judgments was then evaluated using the conventional AHP procedure. The calculated values of the maximum eigenvalue, consistency

index, and consistency ratio demonstrate a high level of consistency among the pairwise comparisons. Since the CR value is well below the acceptable threshold of 0.10, the judgments are considered reliable. Furthermore, the close agreement between the FAHP and AHP weights indicates that the ranking of criteria remains stable even after incorporating uncertainty into the evaluation process.

Table 5. Crisp Analytic Hierarchy Process (AHP) pairwise comparison matrix (n = 5)

Criterion	C1	C2	C3	C4	C5	Weight
C1	1	3	5	4	5	0.482
C2	0.33	1	2	2	3	0.203
C3	0.2	0.5	1	0.5	2	0.099
C4	0.25	0.5	2	1	3	0.150
C5	0.2	0.33	0.5	0.33	1	0.065
$\lambda_{\max} = 5.13$; $RI = 1.12$; $CI = 0.03$; $CR = 0.027$						

Note: n = Sample size; RI = Random Index; CI = Consistency Index; CR = Consistency Ratio.

4.7 Capacity Calculations and Scenario Development

The remaining capacity values used in this study were derived from detailed lagoon geometry surveys rather than assumptions. Bathymetric surveys of Cells 4E, 4W, and 5 were conducted in 2023 and 2025 using a SeaFloor Hydron system, complemented by a Light Detection and Ranging (LiDAR) drone survey to capture point-cloud and coordinate data. The datasets obtained from both survey methods were post-processed using DJI Terra and CloudCompare software and subsequently integrated as surface models in Civil 3D. The processed surface data were then exported to Microsoft Excel, where lagoon volumes were calculated using the average end-area method to develop stage–area curves. The resulting remaining capacities are presented in Table 6.

Table 6. Capacity scenarios and normalized values

Alternative	2025 Measured Capacity (m ³)	Baseline Normalized	Capital Add-on (m ³)	Post-project Effective Capacity (m ³)	Post-project Normalized
Cell 4E	1,917,218	1.00	0	1,917,218	1.00
Cell 4W	649,997	0.34	0	649,997	0.34
Cell 5	286,898	0.15	1,400,000	1,686,898	0.88

Based on these analyses, the baseline remaining capacities of Cells 4E, 4W, and 5 were estimated to be 1,917,218 m³, 649,997 m³, and 286,898 m³, respectively. Among the three alternatives, Cell 4E exhibited the largest available capacity, indicating the longest potential disposal lifespan. Cell 4W possessed a substantially lower capacity than Cell 4E but remained a viable alternative due to its moderate storage volume. In contrast, Cell 5 had the smallest remaining capacity under baseline conditions, making it the least favorable option from a storage perspective.

To account for future infrastructure development, an additional post-project scenario was also evaluated. Under this scenario, a proposed capital project would increase the storage capacity of Cell 5 by approximately 1,400,000 m³, resulting in a total effective capacity of 1,686,898 m³. Incorporating this scenario enabled the analysis to capture the potential impact of planned facility upgrades on future disposal-cell suitability rather than relying solely on current operating conditions.

Prior to inclusion in the TOPSIS decision matrix, the capacity values were normalized to ensure comparability with the remaining evaluation criteria, several of which were qualitative in nature. This normalization process allowed capacity to be evaluated alongside operational and engineering considerations within a unified MCDM framework.

4.8 Development of the Qualitative Decision Matrix

The qualitative scores presented in Table 7 for hydraulic connectivity, discharge accessibility, recirculation impact, and operational flexibility were developed through engineering assessment informed by consultations with experienced plant personnel and a detailed understanding of the PRPS lagoon system. Unlike remaining capacity, these criteria cannot be represented by a single measurable quantitative value. Consequently, a relative scoring scale ranging from 0 to 1 was adopted, where 1.0 denotes the most suitable alternative for a given criterion, 0.8 represents a favorable but not optimal alternative, and 0.6 indicates an alternative with comparatively lower suitability.

Table 7. Qualitative alternative scores

Alternative	C2 Hydraulic	C3 Discharge	C4 Recirculation	C5 Flexibility
Cell 4E	0.8	1	0.6	0.6
Cell 4W	1	0.8	0.8	0.6
Cell 5	0.8	1	0.8	0.8

For hydraulic connectivity, Cell 4W received the highest score because it is the most effectively integrated within the existing lagoon flow network, facilitating efficient hydraulic performance. Cells 4E and 5 were assigned scores of 0.8, reflecting their adequate hydraulic integration while being somewhat less favorable than Cell 4W.

With respect to discharge accessibility, Cells 4E and 5 were considered the most advantageous alternatives due to their favorable conditions for practical ash slurry discharge operations. Cell 4W was assigned a score of 0.8, indicating that it remains a feasible disposal option but presents relatively greater operational challenges for discharge.

For the recirculation impact criterion, Cells 4W and 5 were rated more favorably because they are expected to better support downstream water movement and maintain effective recirculation performance within the lagoon system. Cell 4E received a lower score, as its use for disposal could have a comparatively less desirable effect on the closed-loop water return process, despite remaining operationally viable.

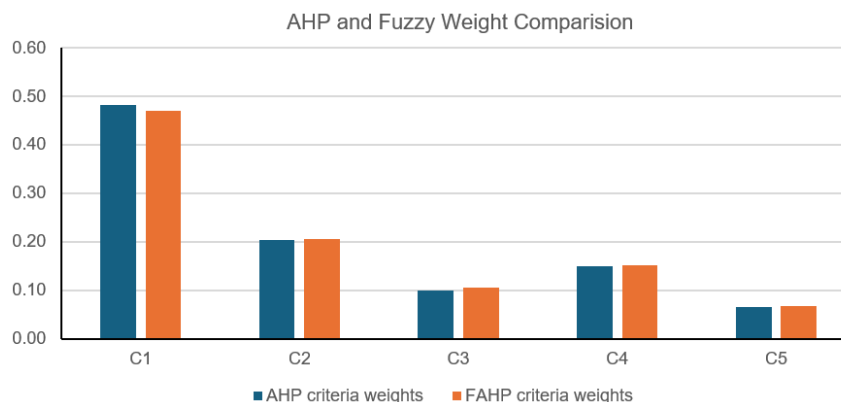
Regarding operational flexibility, Cell 5 was identified as the most favorable alternative, particularly under the post-project scenario, due to its enhanced capacity and ability to support more adaptable future operations. Cells 4E and 4W were assigned lower scores because, although both remain practical disposal options, they provide comparatively less flexibility for long-term operational planning.

Overall, the qualitative scores were not assigned arbitrarily but were derived from a structured engineering evaluation of the physical configuration, hydraulic behavior, and operational characteristics of the PRPS lagoon system. This approach ensured that the assessment reflected actual system functionality and practical operational considerations.

5 Results

5.1 Criteria Weight Distribution

The criterion weights derived from the AHP and FAHP models exhibit a broadly consistent pattern, as illustrated in Figure 2. In both approaches, remaining capacity emerged as the most influential criterion, highlighting its critical role in determining the long-term viability of a disposal cell. Hydraulic connectivity, discharge accessibility, and recirculation impact were identified as secondary factors, although their relative ordering varied slightly between the two methods. Operational flexibility received the lowest weight in both models, indicating that while it remains relevant to decision-making, it is less influential than the core storage and hydraulic performance criteria.

**Figure 2.** Comparison chart of Analytic Hierarchy Process (AHP) and fuzzy weights

The observed weighting structure is consistent with the operational objectives of ash lagoon management. A disposal cell with insufficient remaining capacity would have limited practical value regardless of its performance in other areas. Consequently, the dominance of remaining capacity in both models reflects the fundamental requirement for sustainable ash disposal operations.

One of the primary purposes of FAHP is to account for uncertainty and imprecision in expert judgments and to evaluate their impact on criterion priorities. The results indicate that incorporating fuzzy uncertainty had only a

marginal effect on the weighting structure. Although the FAHP model slightly increased the weights associated with hydraulic connectivity, discharge accessibility, recirculation impact, and operational flexibility, it only marginally reduced the weight assigned to remaining capacity. Importantly, the overall ranking of criteria remained unchanged.

This stability is significant for two reasons. First, it suggests that the original pairwise comparisons were reasonably consistent and robust, as the introduction of fuzzy uncertainty did not substantially alter the resulting weights. Second, it demonstrates that uncertainty in expert judgments does not materially affect the relative importance of the evaluation criteria. The preservation of the priority structure across both methods enhances confidence in the reliability of the weighting process and the subsequent ranking results.

Given that the proposed decision-support framework is based on the FAHP-TOPSIS approach, the FAHP-derived weights were adopted for the final evaluation of lagoon-cell alternatives. The AHP weights are presented primarily for comparative purposes and to illustrate the influence of incorporating fuzzy uncertainty into the weighting process.

5.2 Comparative Lagoon Rankings

The FAHP weights were used in the final TOPSIS analysis, whereas the crisp AHP weights were retained solely as a reference for validation and comparison. The resulting TOPSIS rankings for all evaluated scenarios are presented in Table 8.

Table 8. Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) ranking results for all cases

Scenario	Weighting Method	Alternative	Positive Separation S^+	Negative Separation S^-	Closeness Coefficient	Rank
Baseline	AHP	4E	0.036	0.385	0.911	1
		4W	0.299	0.093	0.238	2
		5	0.385	0.029	0.069	3
Post-project	AHP	4E	0.037	0.232	0.861	1
		5	0.050	0.192	0.793	2
		4W	0.232	0.036	0.133	3
Baseline	FAHP	4E	0.038	0.376	0.908	1
		4W	0.292	0.092	0.239	2
		5	0.377	0.029	0.072	3
Post-project	FAHP	4E	0.038	0.227	0.857	1
		5	0.049	0.188	0.791	2
		4W	0.227	0.036	0.136	3

Note: FAHP = Fuzzy Analytic Hierarchy Process; AHP = Analytic Hierarchy Process.

The ranking results demonstrate a high degree of consistency across all evaluated scenarios. Cell 4E emerged as the highest-ranked alternative under both baseline and post-project conditions, regardless of whether AHP- or FAHP-derived weights were used. This finding is significant because it indicates that the superiority of Cell 4E is not dependent on a particular weighting approach. Rather, it reflects the cell's strong overall performance across the full set of evaluation criteria, making it the most suitable disposal alternative from a system-wide perspective.

The most notable change in ranking occurred between Cells 4W and 5. Under baseline conditions, Cell 4W ranked above Cell 5, primarily due to the latter's limited remaining capacity. With a normalized capacity value of only 0.15, Cell 5 was disadvantaged despite its favorable performance in several qualitative criteria. However, following the inclusion of the proposed capital expansion project, the effective capacity of Cell 5 increased substantially. As remaining capacity was identified as the most influential evaluation criterion, this increase significantly improved the overall performance of Cell 5, allowing it to surpass Cell 4W and attain the second-ranked position in the post-project scenario.

These results highlight the importance of incorporating future infrastructure developments into disposal planning. While Cell 4W remains a technically viable option because of its favorable hydraulic characteristics, the capacity enhancement project fundamentally changes the competitive position of Cell 5 and improves its long-term suitability for ash slurry disposal.

Although AHP-TOPSIS rankings are presented for comparison purposes, the primary decision-support framework adopted in this study is the FAHP-TOPSIS approach. Since FAHP was selected as the principal weighting method to account for uncertainty in expert judgments, the FAHP-TOPSIS results are considered the primary basis for decision-making. A comparative summary of the rankings obtained from all evaluated scenarios is presented in Table 9.

Table 9. Final comparative ranking summary

Scenario	AHP Ranking	FAHP Ranking	Preferred Cell	Interpretive Comment
Baseline	4E > 4W > 5	4E > 4W > 5	4E	Largest current capacity outweighs lower recirculation score
Post Project	4E > 5 > 4W	4E > 5 > 4W	4E	Cell 5 becomes competitive once capacity is increased

Note: FAHP = Fuzzy Analytic Hierarchy Process; AHP = Analytic Hierarchy Process; 4E = Cell 4E; 4W = Cell 4W.

5.3 Sensitivity Analysis of Results

A sensitivity analysis was conducted to assess the robustness and reliability of the post-project FAHP-TOPSIS ranking results by examining the effect of variations in criterion weights (Table 10). Since remaining capacity (C1) received the highest weight in the FAHP framework and was identified as the most influential criterion, a one-factor-at-a-time sensitivity analysis was performed by varying its weight by $\pm 10\%$. The weights of the remaining criteria were proportionally adjusted to maintain normalization, and the TOPSIS procedure was repeated using the same post-project decision matrix.

Table 10. Sensitivity analysis of Fuzzy Analytic Hierarchy Process–Technique for Order Preference by Similarity to Ideal Solution (FAHP-TOPSIS) results

Case	Closeness Coefficient			Ranking
	Cell 4E	Cell 5	Cell 4W	
Base	0.861	0.793	0.133	4E > 5 > 4W
$C_1 + 10\%$	0.864	0.792	0.135	4E > 5 > 4W
$C_1 - 10\%$	0.863	0.794	0.131	4E > 5 > 4W

The results indicate that the ranking of the alternatives remained unchanged across all sensitivity scenarios. In each case, Cell 4E retained the highest ranking, followed by Cell 5 and Cell 4W in second and third positions, respectively. Although minor variations were observed in the closeness coefficient values, these changes were insufficient to alter the relative ordering of the alternatives.

The stability of the rankings demonstrates that the proposed decision model is robust to moderate variations in the weight assigned to the most critical criterion. This finding provides additional confidence in the reliability of the recommended alternative, indicating that the selection of Cell 4E is not dependent on a single set of criterion weights. Rather, the preferred ranking is maintained even when reasonable uncertainty is introduced into the weighting structure, thereby supporting the validity and practical applicability of the FAHP-TOPSIS framework for lagoon-cell selection.

6 Discussion

6.1 Interpretation of Ranking Results

The results indicate that Cell 4E is the most suitable alternative for future ash slurry disposal based on the evaluation criteria considered in this study. This conclusion is supported from both analytical and engineering perspectives. From an analytical standpoint, Cell 4E consistently achieved the highest closeness coefficient across all evaluated scenarios, including both baseline and post-project conditions and under both AHP- and FAHP-derived weighting schemes. The consistency of this outcome demonstrates the robustness of its overall performance relative to the competing alternatives.

From an engineering perspective, Cell 4E combines the largest available disposal capacity with excellent discharge accessibility, making it a practical and effective option for immediate operational use. Although its recirculation performance is less favorable than that of Cells 4W and 5, this limitation is offset by its superior performance in the more heavily weighted criteria, particularly remaining capacity. This outcome is consistent with the theoretical foundation of the TOPSIS methodology, whereby an alternative that is simultaneously closer to the positive ideal solution and farther from the negative ideal solution across multiple weighted criteria is expected to maintain its ranking superiority even under varying decision contexts [6, 19].

Another important finding relates to the performance of Cell 5 under different planning scenarios. Under baseline conditions, Cell 5 was the least attractive alternative due to its limited remaining capacity, which significantly reduced its overall suitability. However, when the proposed capacity expansion project was incorporated into the analysis, the ranking of Cell 5 improved substantially, elevating it to the second position. This improvement highlights the

significant influence of storage capacity on disposal-cell selection and demonstrates the potential benefits of future infrastructure investments.

From a planning perspective, this finding is particularly valuable because it reveals that the proposed capital project contributes more than simply increasing storage volume. By substantially improving the overall suitability of Cell 5, the project enhances operational flexibility and provides an additional viable disposal option for future ash management activities. Consequently, the analysis offers important insights for long-term lagoon planning by quantifying the strategic value of infrastructure development within an integrated decision-making framework.

6.2 Influence of Capacity and Scenario Choice

The prominence of remaining capacity within the decision framework is consistent with the fundamental operational objective of the ash lagoon system. A disposal cell with insufficient storage capacity cannot support long-term ash placement, regardless of its performance with respect to hydraulic or operational characteristics. Consequently, remaining capacity emerged as the most influential criterion in both the AHP and FAHP analyses, accounting for approximately half of the total criterion weight. This result reflects the practical reality that disposal longevity is a primary consideration in lagoon-cell selection.

At the same time, the findings demonstrate the importance of incorporating multiple planning scenarios into the evaluation process. If only existing capacity conditions were considered, Cell 5 would be viewed as a relatively unattractive alternative due to its limited available storage volume. Conversely, an assessment based solely on post-project conditions could overlook the operational constraints associated with its current state. By evaluating both baseline and post-project scenarios, the proposed framework captures the dynamic nature of the decision problem and provides a more comprehensive assessment of alternative suitability over time.

This scenario-based approach represents a key strength of the study, as it acknowledges that infrastructure planning decisions are inherently influenced by future developments and changing operational conditions. Rather than producing a static recommendation, the framework evaluates how alternative rankings evolve in response to planned capacity enhancements. Such an approach is consistent with established MCDM practices, which recognize that different decision contexts may yield different yet equally rational rankings without compromising the validity of the underlying methodology. As a result, the analysis provides a more realistic basis for long-term disposal planning and infrastructure investment decisions.

6.3 Practical Implications of the Study

From a practical planning perspective, the results provide two important insights for ash lagoon management. In the short term, under existing operating conditions, Cell 4E represents the most suitable disposal alternative due to its superior overall performance across the evaluation criteria. Its substantial remaining capacity and favorable discharge accessibility make it the most rational choice for immediate operational implementation. In the medium term, however, Cell 5 emerges as a viable secondary option, particularly if the proposed capacity enhancement project is successfully completed. The improved ranking of Cell 5 under post-project conditions highlights its potential strategic value and suggests that future infrastructure investments can significantly influence disposal planning decisions. Consequently, the ranking results can support both current operational decision-making and long-term planning initiatives.

Another practical contribution of the proposed framework is its ability to provide a transparent and defensible basis for decision-making. Rather than relying on a subjective recommendation that a particular cell is preferable, the framework systematically evaluates each alternative against a clearly defined set of engineering and operational criteria. The resulting rankings, together with the associated weighting structure and performance scores, offer a comprehensive explanation of why a particular alternative is preferred. This transparency facilitates communication among plant management, engineers, operations personnel, and external consultants by providing a common analytical basis for discussion. As a result, the framework not only supports the selection of disposal alternatives but also enhances stakeholder confidence in the decision-making process through a clear and evidence-based justification of the results.

6.4 Limitations of the Study

Several limitations of this study should be acknowledged. First, some evaluation criteria, including hydraulic connectivity, discharge accessibility, recirculation impact, and operational flexibility, are qualitative in nature and were assessed using expert engineering judgment rather than direct quantitative measurements. Although these assessments were informed by consultations with experienced plant personnel and operational knowledge of the PRPS lagoon system, they remain subjective and should be interpreted accordingly.

Second, the analysis was limited to three alternatives—Cells 4E, 4W, and 5—as these represent the disposal cells currently considered operationally feasible within the PRPS lagoon system. The objective was to evaluate realistic disposal options rather than all possible lagoon areas.

Third, the framework was designed primarily to assess the operational suitability of lagoon-cell alternatives. Therefore, the selected criteria focused on key engineering and operational factors, including remaining capacity, hydraulic connectivity, discharge accessibility, recirculation impact, and operational flexibility. Environmental factors were not included as separate criteria because all alternatives are located within the same lagoon system and operate under a common environmental management framework.

Future studies could extend the framework by incorporating environmental criteria such as seepage and leachate risk, groundwater contamination potential, dust generation, and ecological sensitivity where relevant data are available. This would enable a more comprehensive evaluation of lagoon-cell alternatives from both operational and environmental perspectives.

6.5 Recommendations

Based on the results of the analysis, Cell 4E is recommended as the preferred ash disposal alternative for current operations, as it consistently achieved the highest ranking across all evaluated scenarios. Its combination of high remaining capacity and favorable operational characteristics makes it the most suitable option for immediate implementation. Cell 5 is recommended as a strategic future alternative, provided that the proposed capacity expansion project is successfully completed. In such a case, the decision model should be updated using the final capacity estimates and any revised operational constraints, followed by a re-evaluation of the alternatives.

The proposed FAHP–TOPSIS framework should be retained as a reusable decision-support tool for future lagoon management and planning. Its flexible structure allows the incorporation of updated capacity information, changing operational requirements, and additional disposal alternatives without modifying the underlying methodology.

Future research could extend the framework by incorporating environmental, economic, and maintenance-related criteria, including seepage and groundwater contamination risks, dust control, ecological sensitivity, infrastructure modification costs, and long-term monitoring requirements. In the longer term, the development of a new disposal cell west of the existing lagoon system may be considered as a potential expansion strategy to increase disposal capacity. However, such an option would require further evaluation of land availability, hydraulic connectivity, environmental impacts, operational feasibility, and capital investment requirements before implementation.

7 Conclusion

This study developed and applied an integrated FAHP–TOPSIS framework to support the selection of the most suitable ash lagoon cell at PRPS. Five evaluation criteria were considered: remaining capacity, hydraulic connectivity, discharge accessibility, recirculation impact, and operational flexibility. A crisp AHP consistency analysis was first conducted to verify the reliability of expert judgments, after which FAHP was used to account for uncertainty in the pairwise comparisons. The TOPSIS method was then employed to rank three feasible lagoon-cell alternatives under both baseline and post-project capacity scenarios.

The results consistently identified Cell 4E as the preferred alternative across all evaluated cases, indicating its overall superiority in terms of both capacity and operational performance. Cell 5 exhibited limited suitability under baseline conditions due to its low remaining capacity; however, its ranking improved substantially following the proposed capacity expansion project, making it the second most favorable option. Although Cell 4W demonstrated strong hydraulic characteristics, its comparatively lower capacity reduced its overall suitability and limited its final ranking. Overall, the findings demonstrate the effectiveness of the proposed FAHP–TOPSIS framework as a practical decision-support tool for lagoon-cell selection and long-term ash disposal planning.

Author Contributions

Conceptualization, S.H. and G.K.; methodology, S.H., G.K., and S.M.; software, S.H. and G.K.; validation, S.H. and G.K.; formal analysis, S.H.; investigation, S.H. and G.K.; resources, G.K.; data curation, S.H.; writing-original draft preparation, S.H.; writing-review and editing, G.K. and S.M.; visualization, S.H.; supervision, G.K.; project administration, G.K. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this work, the authors used Grammarly Premium and/or ChatGPT to improve the quality of the writing and check for any grammatical errors. After using these tools or services, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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